



C23-M-404

23448

BOARD DIPLOMA EXAMINATION, (C-23)
MARCH/APRIL—2025
DME – FOURTH SEMESTER EXAMINATION
ENGINEERING MATERIALS

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **three** marks.
(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Draw stress-strain diagram for ductile material and indicate salient points on it. 3
2. State the principle of radiography test. 2+1
3. List out of engineering applications for plastics across different fields. (Any six points) 3
4. What are the limitations of powder metallurgy? 3
5. What are the byproducts of blast furnace and what purpose they are used? 3
6. State the advantages of steel making in L-D process. (Any three points) 3
7. Distinguish between interstitial and substitutional solid solutions. 3
8. State the importance of heat treatment of steel. ($\frac{1}{2}$ for each method) 3
9. Write down the properties and applications of HSS of stainless steel. $1\frac{1}{2}+1\frac{1}{2}$
10. What are the desired properties of bearing metals? 3

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PART—B

10×5=50

Instructions : (1) Answer *any five* questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

- 11.** Explain Charpy test procedure with line and specimen diagram. What are the factors promoting the grain size? What is the effect of grain size on mechanical properties? 10
- 12.** Explain Ultrasonic testing with a neat sketch. 5+5
- 13.** Describe transfer moulding process of plastics with a neat sketch. 5+5
- 14.** Explain steps involved in the production process of powder metallurgy. 10
- 15.** Explain with neat sketch how pig iron is produced in blast furnace. 5+5
- 16.** (a) Compare between acid open hearth process and basic open hearth process. 5
(b) Write differences between annealing and normalizing. 5
- 17.** Explain the following heat treatment processes : $2\frac{1}{2}+2\frac{1}{2}+2\frac{1}{2}+2\frac{1}{2}$
(a) Normalizing
(b) Hardening
(c) Nitriding
(d) Cyaniding
- 18.** Write down the composition, properties and uses of nonferrous alloys. 3+4+3

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